Calculating single-electron states in quantum dots

Julia Potempa

AGH University of Krakow

April 28, 2026

Contents

1 Task 1 - Visualization of Basis Functions **2**

2 Task 2 - Matrix Elements of S, K, and V **3**

3 Task 3 - Eigenstates for $\hbar\omega_x = 80$ meV, $\hbar\omega_y = 200$ meV **3**

4 Task 4 - Energy Dependence on $\hbar\omega_x$ **4**

5 Task 5 - States in the Quasi-1D Regime and Comparison with Experiment **5**

Objective

The aim of this exercise was to numerically determine the energy levels and wavefunctions of an electron confined in a two-dimensional harmonic potential (quantum dot model) using the Galerkin method with a Gaussian basis. The 2D Hamiltonian of the system is:

$$\hat{H} = -\frac{1}{2m^*}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + \frac{1}{2}m^*(\omega_x^2 x^2 + \omega_y^2 y^2), \tag{1}$$

where $m^* = 0.24\, m_e$ is the effective electron mass and ω_x, ω_y are the oscillator frequencies along x and y, respectively. The basis consists of Gaussian wavefunctions $\phi_k(x, y)$ placed on a regular $n \times n$ grid of nodes (here $n = 9$, giving $N = 81$ basis functions). The problem reduces to a generalized eigenvalue problem $H\mathbf{c} = E\, S\mathbf{c}$, where H and S are the Hamiltonian and overlap matrices. The model and parameters are motivated by the experimental study of self-assembled InAs quantum dots embedded in an AlAs matrix by Teichmann et al. [1], in which up to five equidistant energy levels with a spacing of 80 meV and an effective oscillator mass $m^* = 0.24\, m_0$ were resolved by cross-sectional scanning tunneling spectroscopy (STS).

1 Task 1 - Visualization of Basis Functions

Description

The basis functions $\phi_k(x, y)$ are products of Gaussians along the x and y axes:

$$\phi_k(x, y) = (\alpha_x \pi)^{-1/4}\, e^{-(x-x_k)^2/(2\alpha_x)} \cdot (\alpha_y \pi)^{-1/4}\, e^{-(y-y_k)^2/(2\alpha_y)}, \tag{2}$$

where $\alpha_{x,y} = \frac{1}{m^* \omega_{x,y}}$ (in atomic units) are the width parameters and (x_k, y_k) are the coordinates of the k-th grid node. The nodes are placed on a 9×9 grid with spacing $\Delta x = 1$ nm spanning $[-a, a]$ in both directions, where $a = \Delta x \cdot (n-1)/2 = 4$ nm.

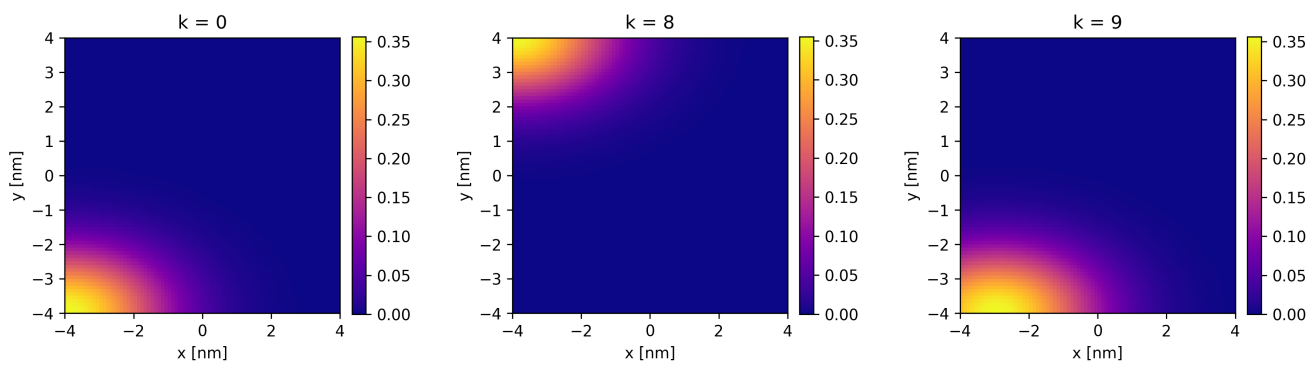

Figure 1: Gaussian basis functions $|\phi_k(x, y)|$ for selected grid nodes: $k = 0$ (bottom-left corner), $k = 8$ (upper-left corner), and $k = 9$ (lower-left corner of the second column). The anisotropy of the Gaussians is clearly visible - the extent along y is smaller than along x, consistent with the stronger confinement frequency $\omega_y > \omega_x$.

2 Task 2 - Matrix Elements of S, K, and V

Description

The overlap, kinetic energy, and potential energy matrix elements are computed analytically:

$$S_{kl} = e^{-(x_k - x_l)^2/(4\alpha_x) - (y_k - y_l)^2/(4\alpha_y)}, \tag{3}$$

$$K_{kl} = -\frac{1}{2m^*}\left[\frac{(x_k - x_l)^2 - 2\alpha_x}{4\alpha_x^2} + \frac{(y_k - y_l)^2 - 2\alpha_y}{4\alpha_y^2}\right] S_{kl}, \tag{4}$$

$$V_{kl} = \frac{m^*}{2}\left[\frac{\omega_x^2\left((x_k + x_l)^2 + 2\alpha_x\right)}{4} + \frac{\omega_y^2\left((y_k + y_l)^2 + 2\alpha_y\right)}{4}\right] S_{kl}. \tag{5}$$

All calculations are performed in atomic units (a.u.), with positions converted from nanometres to Bohr radii ($a_0 = 0.0529$ nm) and energies expressed in Hartree ($E_h = 27211.6$ meV).

3 Task 3 - Eigenstates for $\hbar\omega_x = 80$ meV, $\hbar\omega_y = 200$ meV

Description

The generalized eigenvalue problem $H\mathbf{c} = E\,S\mathbf{c}$ was solved numerically using `scipy.linalg.eigh`. The resulting wavefunctions:

$$\Psi_n(x, y) = \sum_k c_k^{(n)} \phi_k(x, y) \tag{6}$$

are presented as probability densities $|\Psi_n|^2$.

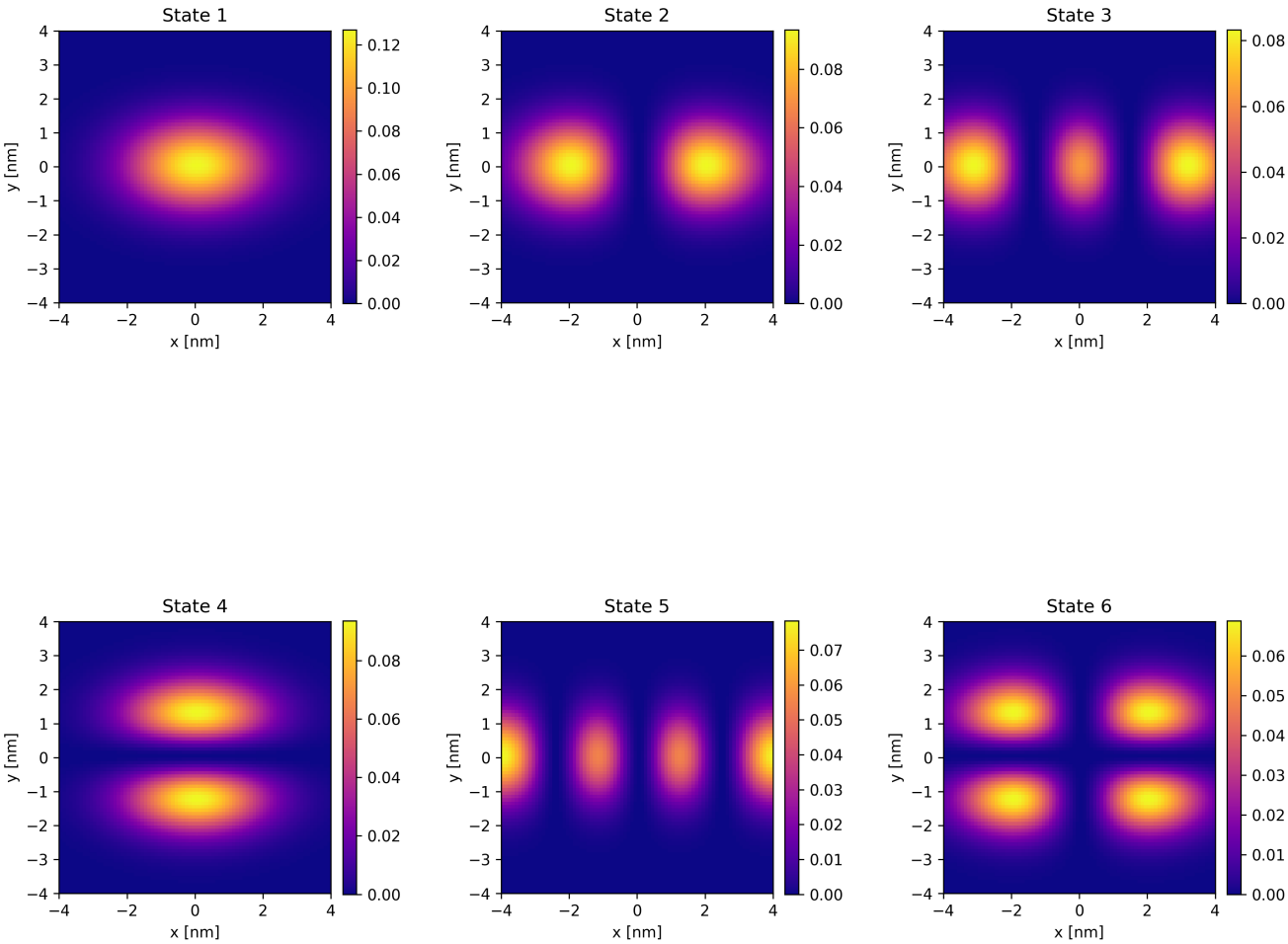

Figure 2: Probability densities $|\Psi_n(x,y)|^2$ for the six lowest eigenstates at $\hbar\omega_x = 80$ meV and $\hbar\omega_y = 200$ meV. The states correspond to combinations of quantum numbers (n_x, n_y): state 1 - $(0,0)$, state 2 - $(1,0)$, state 3 - $(0,1)$, state 4 - $(2,0)$, state 5 - $(1,1)$, state 6 - $(3,0)$.

4 Task 4 - Energy Dependence on $\hbar\omega_x$

Description

The variation of the ten lowest energy levels as a function of $\hbar\omega_x \in [10, 500]$ meV was investigated at fixed $\hbar\omega_y = 200$ meV. The analytical curves:

$$E_{n_x,n_y} = \hbar\omega_x\left(n_x + \tfrac{1}{2}\right) + \hbar\omega_y\left(n_y + \tfrac{1}{2}\right) \tag{7}$$

for several (n_x, n_y) combinations are overlaid as dashed lines.

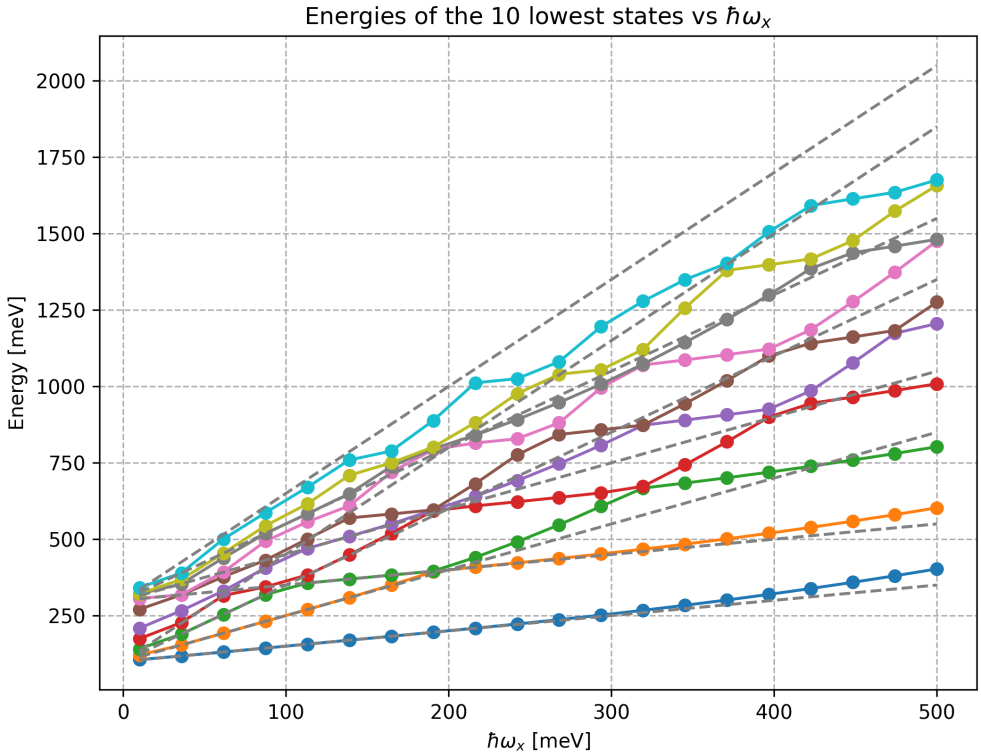

Figure 3: Energies of the 10 lowest states as a function of $\hbar\omega_x$ (points - numerical results, dashed lines - analytical values). A linear increase in the energies of states excited along x is visible. The apparent sudden jumps on some numerical curves near level crossings are an artefact of state reordering: as ω_x increases, states with higher n_x rise in energy and interchange their ordering with states having $n_y > 0$. This effect is most pronounced near $\hbar\omega_x \approx 200$ meV, where the system approaches the isotropic limit $\omega_x = \omega_y$.

5 Task 5 - States in the Quasi-1D Regime and Comparison with Experiment

Description

Task 5 connects the numerical model directly to the experimental results of Teichmann et al. [1]. In that work, cross-sectional STS on a single InAs quantum dot embedded in an AlAs matrix revealed up to five equidistant energy levels with a spacing of 80 meV and wavefunction nodal structures identical to those of a 1D harmonic oscillator. This behaviour arises because the cleavage plane breaks the in-plane rotational symmetry of the QD, leaving effective 1D confinement along the [110] direction dominant in the accessible bias range.

To reproduce this quasi-1D regime numerically, the confinement in the y direction was increased to $\hbar\omega_y = 400$ meV while keeping $\hbar\omega_x = 80$ meV. Under these conditions the large y-quantization energy pushes all $n_y > 0$ states well above the first five levels, so the lowest eigenstates are excited

exclusively along x.

Eigenenergies are given by:

$$E_{n_x,n_y} = \hbar\omega_x\left(n_x + \tfrac{1}{2}\right) + \hbar\omega_y\left(n_y + \tfrac{1}{2}\right). \tag{8}$$

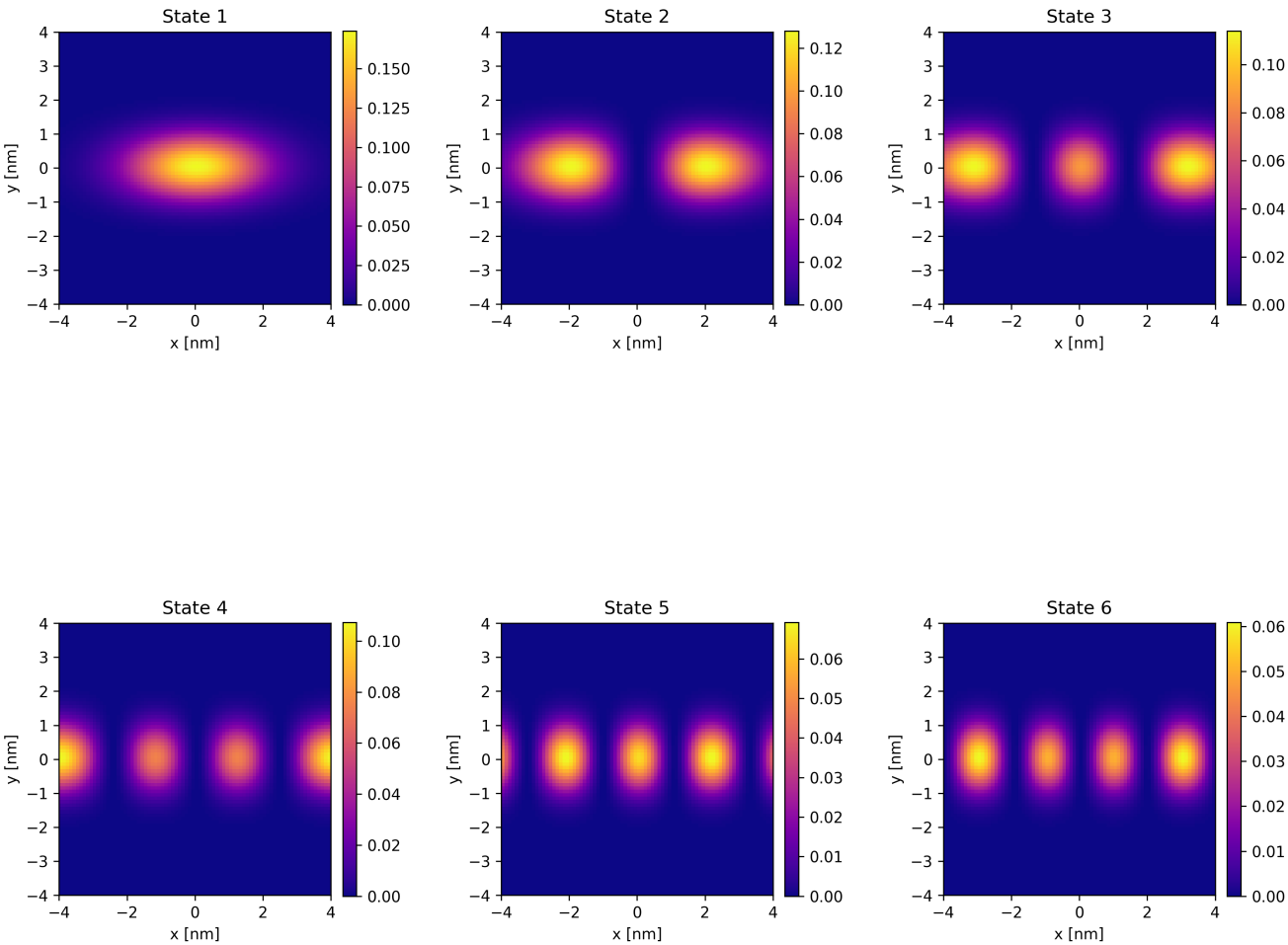

Figure 4: Probability densities $|\Psi_n|^2$ for the six lowest states at $\hbar\omega_x = 80$ meV, $\hbar\omega_y = 400$ meV. The first five states ($n_x = 0, 1, 2, 3, 4$, $n_y = 0$) are excited exclusively along x, confirming the quasi-1D character of the system. The sixth state, where y-excitation first appears ($n_y = 1$), lies significantly higher in energy. The nodal structure - zero nodes for $n_x = 0$, one node for $n_x = 1$, etc. -matches directly the STM dI/dV maps reported in Fig. 2 of Ref. [**teichmann2013**].

Summary

The Galerkin method with an anisotropic Gaussian basis enabled efficient numerical determination of energy levels and wavefunctions of an electron in a 2D harmonic quantum dot. The results are in excellent agreement with the analytical solutions of the 2D harmonic oscillator across all investigated parameter regimes. A 9×9 node grid proved sufficient to accurately describe the lowest-lying states. The analysis as a function of ω_x revealed level crossings and degeneracies in the isotropic limit $\omega_x = \omega_y$. Most importantly, tuning $\hbar\omega_y$ to 400 meV reproduced the quasi-1D harmonic oscillator wavefunction sequence observed by Teichmann et al. in STM measurements on InAs/AlAs quantum dots, confirming that the simple harmonic model with an enhanced effective mass captures the essential physics of these nanostructures.

References

[1] K. Teichmann, M. Wenderoth, H. Prüser, K. Pierz, H. W. Schumacher, and R. G. Ulbrich, *Harmonic Oscillator Wave Functions of a Self-Assembled InAs Quantum Dot Measured by Scanning Tunneling Microscopy*, Nano Lett. **13**, 3571–3575 (2013).